The following points may be noted regarding the covalent bonds:

1. Covalent bonds are formed by sharing of valence electrons.
2. In the formation of covalent bond, each valence electrons of an atom forms direct bond with the valence electrons of an adjacent atom. For this reason, valence electrons in a semiconductor are not free.

3.6.3 Crystals

A substance in which the atoms or molecules are arranged in orderly pattern is known as a crystal.

All semiconductors have crystalline structure. From the Figure 3.17, it is clear that each atom is surrounded by neighbouring atoms in a repetitive manner; therefore, a piece of Germanium is generally called crystalline structure.

3.6.4 Commonly Used Semiconductor

There are many semiconductors available, but very few of them have practical application in Electronics. The two most frequently used materials are Germanium (Ge) and Silicon (Si). These two are widely used because the energy required to break their covalent bond is very small (i.e., energy required to release an electron from their valence bonds) being about 0.7eV for Germanium and 1.1eV for Silicon. Let us see about these two.

1. **Germanium:** Germanium is the model substance among the semiconductors. The main reason being that it can be purified well and crystallized easily. It is discovered in 1886. It is recovered from the ash of certain coals. The atomic number of germanium is 32, i.e., it has 32 protons and 32 electrons. It is clear that germanium atom has 4 valence electrons i.e., tetravalent element.
2. **Silicon:** Silicon is an element available in most of the common rocks. Actually sand is silicon dioxide. The silicon compounds are chemically reduced to silicon which is 100% pure for use as semiconductors. The atomic number of silicon is 14 and hence it has 14 protons and 14 electrons. It is very clear that silicon atom has four valence electrons i.e. tetravalent element.

With references to the nature and characteristics of semiconductors, there are two types of current flow. As we know, at low temperature the conduction-band in semiconductors will be empty (no free electrons), but increase in temperature causes changes in conduction-band i.e. few valence electron would come to conduction-band thus constituting electric current. Here, we have to note one thing, each time a valence electron enters into the conduction band a hole is created in the valence band. These holes also contribute to current and this current is called as hole current. In fact, hole current is the most significant concept in semiconductors.

3.6.5 Energy Band Description of Semiconductors

It has been already discussed that a semiconductor is substance whose resistivity lies between conductors and insulators. The resistivity is of the order of 10 to 0.5 ohm metre. However, semiconductor can

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